JEEVAN RAGHA SOURYA MUSULURU

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.00
Narayana Junior College	Class XII (BIEAP)	2019	10.00%
Sri Chaitanya English Medium School	Class X (BSEAP)	2017	10.00%

INTERNSHIP

Founding Team Intern | Eknal Technologies Pvt. Ltd.

- **Led market research and product design** for 3 platforms (Eknal Bridge, Hire, Track) to align features with customer needs.
- **Designed user flows and UI screens** for all three apps based directly on customer feedback and requirements.
- **Recruited 28 interns in 2 months** by running hiring campaigns, interviewing candidates, and selecting top talent.
- **Created onboarding documentation** to train new hires and get them up to speed quickly.
- **Managed daily team operations** by assigning tasks, leading catch-up calls, and keeping projects moving forward.

Project Management Intern | Foruppo

- **Incubated Start-up Party**, an invite-only event series offering networking and mental health support for startup founders.
- **Recruited 22 psychologists** in a single day through direct outreach to support founder mental health.
- **Secured free event venues** at co-working spaces in Bangalore, Delhi, and Mumbai through value-pitch negotiations.
- **Managed team workflows and intern hiring** by organizing lead datasets and coordinating with college placement cells.

Mechanical Engineer Intern | Nelcast Pvt. Ltd.

- **Learned the metal casting process** by observing sand mold creation, metal melting, pouring, and cooling.
- **Studied surface finishing methods**, observing sand blasting, shot blasting, and milling.
- **Observed quality inspection techniques** to understand how defects are identified in finished parts.

PROJECTS

Pulsed Laser Annealing (PLA): Uncertainty Quantification, Process Optimization & Phase-Field Modelling

Stage 1 :

- Developed enthalpy-based solver for nano second Rapid pulsed laser annealing of silicon incorporating Beer-Lambert laser energy absorption and heat transfer, using FVM and Newton-Raphson iteration to solve nonlinear heat-transfer PDEs .
- Performed uncertainty quantification via Latin Hypercube Sampling across three published liquid-permittivity datasets, generating 3000 simulations and applying Sobol first-order sensitivity analysis to identify dominant uncertainty sources.
- Discovered that real permittivity (reflectivity) dominates 99.99% of melt-depth uncertainty using Sobol first-order sensitivity indices, contradicting literature assumptions about imaginary permittivity (absorption depth) importance.
- Proposed process window design by optimizing laser-pulse shapes to reduce uncertainty in melt-depth predictions, targeting >95% success rate within 40% tolerance window despite 2.7 x reflectivity variation across permittivity datasets.

Stage2:

- Build a 3-D anisotropic phase-field model in **FEniCS** ,coupling enthalpy-solver output with **grain-growth** PDE, incorporated orientation-dependent interface mobility calibrated via **Kinetic Monte Carlo method with MulSKIPS.**
- **Surrogate model** to predict crystal orientation in [1,1,1] [1,0,0] [1,1,0] directions of silicon.

Statistical Process Control on BOSCH Process Data Set.

- Generated 500-point synthetic dataset for Bosch process. Built **I-MR, X-bar-R** control charts with eight Western

Electric zone rules to detect out-of-control patterns in RF power, bias voltage, chamber pressure, gas flows, and etch rate.
- Root-caused anomalies (in RF matching drift, arc event, polymer buildup, mass flow sticking) using control chart signals.

Strontium Titanate Deposition via Pulsed Laser Deposition (PLD)

- Executed high-vacuum Pulsed Laser Deposition (PLD) of Strontium Titanate thin films on platinized silicon substrate .
- Implemented a multi-step solvent cleaning protocol (Acetone and IPA followed by DI water rinse and blow-drying) to eliminate surface contamination and coffee-ring defects prior to chamber loading.
- Learnt to Configure high-vacuum chamber operations using a combined rotary and turbomolecular pump system down to microbar pressures alongside direct-current-heated thermal ramping at a controlled rate up to high temperatures.

Precipitate Thermo-migration: Phase-Field Simulation

- Constructed **1D-phase-field thermo-migration model** with julia by solving the 4th-order **Cahn-Hilliard equation** with a temperature-dependent double-well free energy.
- Derived and implemented a **semi-implicit Fourier spectral solver**, achieving numerical stability and a **1250x** speedup (80,000 vs 108 timesteps) **over explicit finite difference solver.**
- Built a threshold-based precipitate-tracking algorithm and validated simulation physics through free-energy monotonic decay and composition conservation.

Molecular Dynamics Simulation: Inert Gas Phase Transitions

- Built an **MD engine from scratch in Python** to simulate **atomic trajectories** and inert **gas solid-to-liquid phase transitions.**
- Optimized computational efficiency from **O(N*N) to O(N)** scaling by implementing **Verlet Neighbor Lists** and vectorizing interatomic force routines using NumPy.
- Evaluated structural and dynamical transport properties by computing Radial Distribution Functions and tracking unwrapped coordinates to calculate diffusion coefficients via Mean Square Displacement (MSD).

SKILLS

Fabrication: Pulsed Laser Deposition, LPCVD , SPC.

Characterization: Optical Microscopy, SEM, TEM, X-Ray Diffraction, 4-Probe Station, Hall Co-efficient Measurement.

Programming Languages: Python, Julia, SQL.

Machine Learning: SVD, PCA, ICA, Gaussian Process Regression

Tools: Quantum Espresso, Numpy , Pandas, Matplotlib, Solid works, Auto CAD, TCAD, FeNiCS, Linux, Latex, JMP.

Languages Known: English, Hindi, Telugu, Tamil.

RELEVANT COURSES

Fundamentals of Semiconductor Materials, **Micro and Nanofabrication , Semiconductor Devices,** Fundamentals of Electronic Packaging Materials, **Semiconductor materials and Devices Characterization Lab** , **Imaging** in Materials Science .

POSITON OF RESPONSIBILITY

Placement Co-Ordinator | Office of Career Services, IITH.

- **Pitched 50+ Material Science companies** by matching job opportunities with student skills to drive campus placements.
- **Co-organized Japan Career Day**, helping manage event logistics and connect students with international job opportunities.

Events and Ideation Head | Cycling Club, IITH

- Co-ordinated FIT India Cycle-marathon organised **Ministry of Youth Affairs and Sports**.
- Organised Cycle-marathon for **Nasha Mukt Bharat Abhiyaan** .
- Conducts Sundays on cycle every sunday.

PG Buddy-Sunshine | Counselling Cell , IITH

- **Coordinated wellness workshops**--including clay, sound, and art therapy--to support student, faculty, and staff mental health.
- **Organized monthly evening events** for first-year PG students to foster connections and relieve academic stress.
- **Provided peer counseling to 5 students** after receiving formal mental health and active listening training.

HOBBIES

Cooking, Riding Motorcycles, Cycles, Badminton, Video Games.